

JAKE THOMAS MATHEW

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Education

National Institute of Technology (NIT), Surat

2024 – 2028

Bachelor of Technology in Computer Science Engineering

- **CGPA: 7.52/10**
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networking, Object-Oriented Design (OOD), Discrete Mathematics, Artificial Intelligence, Security.

Technical Skills

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

Languages: C++, C, Python, SQL, JavaScript, TypeScript

Backend: Node.js, Express.js, MongoDB, MySQL, REST APIs, JWT, Socket.io

Frontend: React.js, HTML, CSS, Tailwind CSS, EJS, Framer Motion

Tools & Platforms: Git, GitHub, VS Code, Postman, Firebase, Vite

Work Experience

Founder & Full-Stack Developer – Crossfit (Sports Meetup Startup)

2024

GitHub: CrossFit

Surat, India

- Designed and implemented backend architecture serving 500+ users, achieving <200ms latency under 600+ simulated concurrent requests.
- Reduced API response time from ~900ms to ~350ms (~60% improvement) through query optimization and database indexing.
- Implemented secure JWT-based authentication and authorization system, validated through stress testing.
- Integrated Razorpay payment gateway with server-side validation, processing 50,000+ in transactions with zero reconciliation errors.
- Drove product iteration through structured user feedback loops, facilitating 20+ real-world meetups within 3 months.

Projects

TerraShield – AI-Powered Property Loan Fraud Detection Platform

2026

GitHub: TerraShield

- Architected a 4-layer verification pipeline (document, collateral, financial, cross-document) analyzing 10+ document types for fraud detection.
- Built a FastAPI-based fraud-scoring pipeline implementing 15+ validation checks spanning OCR, metadata/font analysis, error-level analysis (ELA), and signature/seal verification.
- Engineered quantitative financial-risk models using Beneish M-Score, Altman Z-Score, and ITR-vs-P&L cross-validation to flag anomalous filings.
- Optimized end-to-end pipeline (OCR + multi-model checks + report generation) to run in <60 seconds per case, with an explainable-risk-scoring dashboard built in Next.js.

Rabuste – Coffee Marketplace & Social Platform

GitHub: Rabuste

- Built a full-stack platform supporting 300+ users across customer, artist, and admin roles, including a real-time chat system handling 100+ concurrent connections.
- Reduced average backend API response time by ~35% through query and endpoint optimization.

Technical Achievements

- **2nd place, Canara Bank Hackathon** – out of 5,500 competing teams.
- Built and deployed **3+ full-stack, production-grade applications** with scalable backend systems and real-time features.
- Designed systems handling **300+ concurrent users** with a consistent focus on latency reduction and query optimization.

Achievements

- Selected in the internal round of **Smart India Hackathon (SIH)**, ranking among the top 25% of teams (50 out of 200+).
- Runner-up, **Google GWoC (Winter of Code)**.
- Runner-up, college-level **Web Wonders Web Development Competition**.
- Led an **AI workshop on prompt engineering** under ACM's Hour of AI, teaching 150+ students to apply LLMs to real-world use cases.
- **School Topper in Computer Science** (97%), Class 12 CBSE.

Leadership Roles

- **Team Lead** – Mindbend (NIT Surat): Led a team organizing Gujarat's largest techno-managerial festival with 1000+ participants, managing execution and cross-team coordination.
- **Organizer & Lead** – ECell Startup Event: Led planning and execution of entrepreneurship-focused events, coordinating teams, speakers, and logistics.